

Chia-Chi (Carol) Hsu

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Education

National Tsing Hua University (NTHU) Sept. 2023–Dec. 2025

M.A., Special Education (GPA: 4.30/4.30)

Thesis: *Cerebral Lateralization of Language and Face Processing in Children with Autism Spectrum Disorder*

Advisor: Prof. Hsin-Jen Hsu

National Tsing Hua University (NTHU) Sept. 2019–May 2023

B.A., Special Education (GPA: 4.14/4.30)

University of California, Berkeley Aug. 2024–May 2025

Visiting Research Student, Berkeley International Study Program (BISP)

Ministry of Education Academic Talent Development Fellow

Research Interests

Developmental cognitive neuroscience; neural mechanisms of language and reading development in children; individual differences in brain–behavior relationships; multimodal approaches to studying brain development and learning (fMRI, fTCD, EEG/ERP).

Research Experience

Research Assistant — Agile Brain and Cognitive Lab, NTHU 2026–Present

Project: *Memory Encoding and Exercise Intervention* **Advisor:** Prof. Zai-Fu Yao

- Investigate whether acute aerobic exercise alters neural systems supporting memory encoding and recognition in children with obesity.
- Conduct structural and functional MRI analyses using FreeSurfer, FSL, and SPM, including cortical reconstruction, volumetric analyses, preprocessing, and first-level modeling.
- Examine whether exercise-related changes in hippocampal and prefrontal activation are associated with individual differences in memory performance and learning outcomes.

Project: *Mathematical Cognition and Electrophysiology* **Advisor:** Prof. Zai-Fu Yao

- Examined behavioral and posterior P300 responses during mathematical verification in adolescent athletes and controls.
- Conducted EEG/ERP preprocessing and analysis using EEGLAB and ERPLAB.
- Investigated how individual differences in mathematical proficiency account for apparent group differences in neural responses.

Master's Thesis Researcher — NTHU 2023–2025

Project: *Language Lateralization and Reading in ASD* **Advisor:** Prof. Hsin-Jen Hsu

- Examined developmental differences in language and face lateralization across children with ASD, typically developing children, and typically developing adults.
- Designed E-Prime paradigms, conducted functional transcranial Doppler ultrasound (fTCD) recordings, and performed laterality index analyses for all 60 participants.

- Found contrasting developmental patterns of language lateralization, with a leftward tendency in typically developing children and an atypical rightward tendency in children with ASD, while face processing was consistently right-lateralized across groups.

Graduate Research Assistant — Child Language Laboratory, NTHU 2023–2025

Project: *Language Development and Sensorimotor Function in DLD* **Advisor:** Prof. Hsin-Jen Hsu

- Investigated associations between sensorimotor function and language development in Mandarin-speaking children with developmental language disorder (DLD).
- Administered standardized assessments of intelligence, language, reading, and behavioral functioning and assisted with eye-tracking task implementation and data collection.

Independent Researcher — University of California, Berkeley Jan 2025–May 2025

Project: *Language Anxiety and Achievement* **Advisor:** Prof. Joanna Reed

- Investigated relationships between language anxiety and English language achievement among Mandarin-speaking exchange students.
- Independently designed and completed the project, including research design, survey development, participant recruitment, data collection, statistical analysis, and conference presentation.
- Findings indicated that language proficiency and anxiety did not show a simple inverse relationship, highlighting substantial individual variability in second-language learning outcomes.

Undergraduate Research Assistant — NTHU & National Taichung University of Education 2020

Project: *Numerical Cognition and Arithmetic*

Advisor: Prof. Kuo-Long Chen and Prof. Yuan Yuan

- Investigated how numerical representations and problem-solving strategies influence arithmetic processing.
- Independently designed a PsychoPy-based experimental paradigm and collected behavioral data from over 50 university students.
- Observed substantial individual variability in solution strategies, response times, and accuracy despite comparable task outcomes.

Publications

Peer-Reviewed Journal Articles

Hsu, C.-C., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. (2026). Baseline mathematical aptitude attenuates group differences in posterior P300 during arithmetic verification in high school athletes. *Experimental Brain Research*, 244(163). <https://doi.org/10.1007/s00221-026-07366-y>

Manuscripts in Preparation

Hsu, C.-C., Yu, H.-Y., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F.

Posterior P300 differences are more robust during multiplication than function verification.

Manuscript in preparation.

Hsu, C.-C., & Yao, Z.-F.

The Taiwanese Blueprint for Inclusive Aquatics for Children with Medical Dependencies.

Manuscript in preparation.

Conference Presentations

Hsu, C.-C., & Hsu, H.-J. (2026, November). *Atypical lateralization of word processing but not face processing in children with autism spectrum disorder*. Poster presentation, Psychonomic Society Annual Meeting, San Diego, CA, United States.

Hsu, C.-C., Yu, H.-Y., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. (2026, November). *Task-invariant P300 responses in athletes during mathematical tasks*. Poster presentation, Psychonomic Society Annual Meeting, San Diego, CA, United States.

Hsu, C.-C., Hsu, H.-Y., Ding, T.-J., Yao, C.-Y., & Yao, Z.-F. (2026, June). *Enhanced posterior P300 during arithmetic verification reflects differences in mathematical proficiency*. Poster presentation, Summit of Psychological Associations in the Asia-Pacific (APPA), Tainan, Taiwan.

Hsu, C.-C. (2025, November). *Exploring language proficiency and anxiety among Mandarin-speaking exchange students at UC Berkeley*. Poster presentation, International Conference on Educational Innovation (ICEI).

Hsu, C.-C., & Hsu, H.-J. (2023, December). *Brain lateralization of language and visual processing in individuals with and without ASD*. Poster presentation, Asia Pacific Society of Speech, Language and Hearing (APSSLH), Ho Chi Minh City, Vietnam.

Honors and Awards

Elected Member, Phi Tau Phi Scholastic Honor Society	2026
Academic Talent Development Fellowship (UC Berkeley), Ministry of Education	2024
International Conference Travel Grant, Ministry of Education	2023
Undergraduate Research Creativity Award, National Science and Technology Council	2022

Teaching Experience

Teaching Assistant — National Tsing Hua University 2023–2025
Undergraduate Courses

- Teaching Materials and Methods for Children with Special Needs (2 semesters)
- Theory and Practice of Individualized Education Programs (2 semesters)
- Special Education Practicum I & II (3 semesters)

Graduate Course

- Research on Theories and Practice of Special Education: supported instruction in research design, experimental methodology, and laboratory orientation in neurophysiological research methods.

Invited Talks and Workshops

Problem Behaviors Across Disability Categories 2024
Teacher Education Center, National Tsing Hua University

Special Education Lesson Planning and Curriculum Design 2023, 2024
Teacher Education Center, National Tsing Hua University

Introduction and Application of PsychoPy in Experimental Design 2021
Department of Special Education, National Tsing Hua University

Technical Skills

Neuroimaging: fMRI (SPM, FSL, FreeSurfer); EEG/ERP (EEGLAB, ERPLAB); fTCD (DopOSCCI)

Programming and Data Analysis: MATLAB; Python; R; SPSS

Experimental Design: E-Prime; PsychoPy

Educational Measurement and Statistics: Rasch Modeling; Item Response Theory (ConQuest)